

Batch: **Roll No.:**
Experiment / assignment / tutorial No. 05
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

Title: Pulse Amplitude Modulation using IC LF398 (Sample and Hold)

Objectives:

1. To understand process and types of sampling
2. To observe natural and flat-top PAM signal

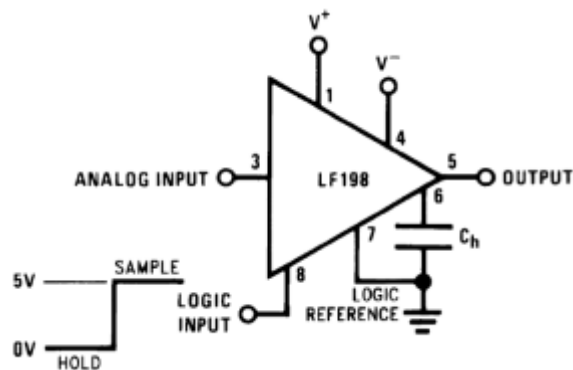
OUTCOME:

1. understand working of analog communication transmitter and receiver systems
2. compare Pulse modulation systems and multiplexing techniques.
3. understand noise effect on the performance of communication systems.

• **Procedure:**

1. Make the connection as shown in figure.
2. Apply the modulating signal amplitude 3-5 Vp-p and the frequency in the range of the 100 to 500Hz.
3. The pulse train amplitude is 5Vpp with frequency in the range of the 1 KHz to 3 KHz.
4. Observe the Sample and hold output at pin no 5 of the IC LF398 with capacitor connected in the circuit.
5. Observe the PAM output at pin no 5 of the IC LF398 with capacitor not connected in the circuit.
6. Plot the waveforms on the graph paper of step 4 and step 5.

• **Functional block diagram of IC LF398:**



• **Circuit diagram: Draw circuit diagram of natural and flat-top PAM.**

• **Procedure:**

1. Make the connection as shown in figure.
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3. The pulse train amplitude is 5Vpp with frequency in the range of the 1 KHz to 3 KHz.
4. Observe the Sample and hold output at pin no 5 of the IC LF398 with capacitor connected in the circuit.

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5. Observe the PAM output at pin no 5 of the IC LF398 after removing capacitor from circuit.

• **Observation Table:**

1) $f_s \geq 2f_m$		2) $f_s < 2f_m$	
Sampling Frequency	Modulating Signal Frequency	Sampling Frequency	Modulating Signal Frequency

- **Show calculations:**
- **Post lab activity:**
 - What is bit depth or resolution in quantization? (1 mark)
 - What role does sampling rate play in digital communication? (1 mark)
 - With only one diagram represent Ideal sampling, natural sampling and flat top sampling. (3 marks)

Signature of faculty in-charge